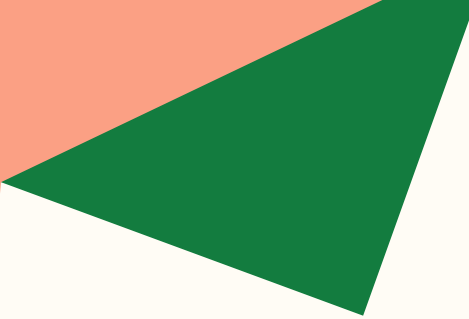


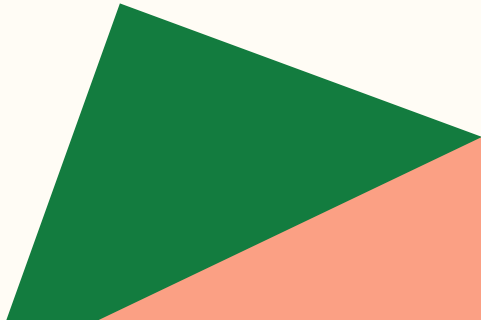


Disaster or Opportunity?: Analyzing the Societal Impacts of AI - RAISE '25

Team ML Super-Man

Sreejato (Sree) Chatterjee, Shubhranshu (Dew) Dutta

University of Rochester



Goal & Data Analysis Framework

Goal: Uncover key themes from news headlines surrounding AI to determine how it will shape the future, derive insights from datasets to develop policy recommendations.

Data Analysis Plan (Python):

1. Data Preprocessing & Exploratory Analysis (n-grams & NER)
2. Sentiment Analysis
3. Topic Modeling

Data Preprocessing & Exploratory Analysis

Datasets Used & Preprocessing Steps

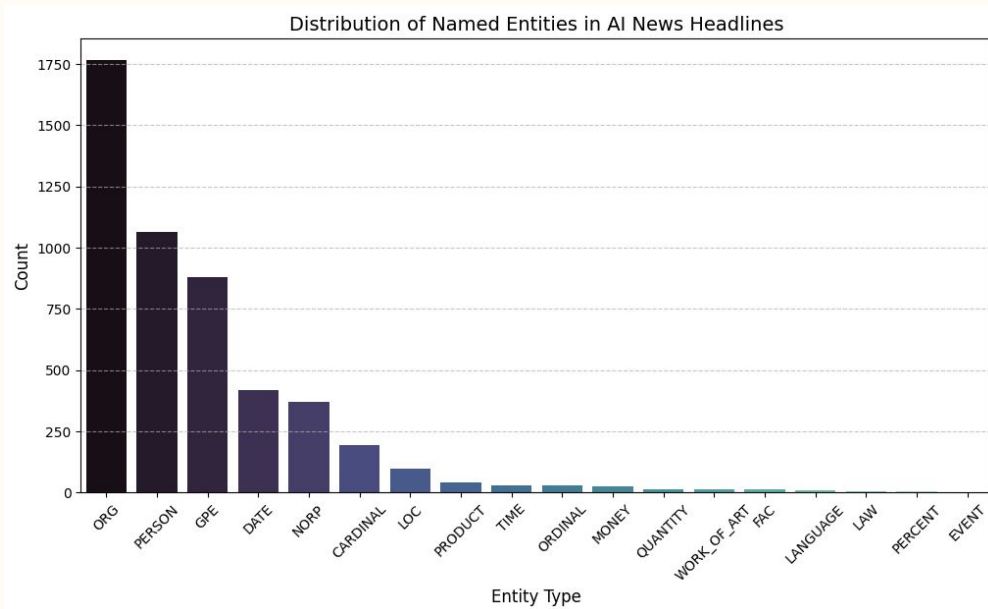
- Combination of 3k, 10k and multilingual datasets
- Multilingual datasets were filtered to only use english news headlines, duplicate headlines dropped, punctuation/symbols removed and all headlines set to lowercase for consistency
- Time series index calculated from extracting time-based features
- Total of 14,013 news headlines analyzed
- For all text-based models, we employed custom stopwords including 'AI', 'ML', 'machine learning', and 'artificial intelligence' to filter out noise and find unique discussions, as well as 'stock' and 'buy' as these words added irrelevance to the data (Samuel et. al 2024)

Named Entity Recognition (NER) & N-Grams

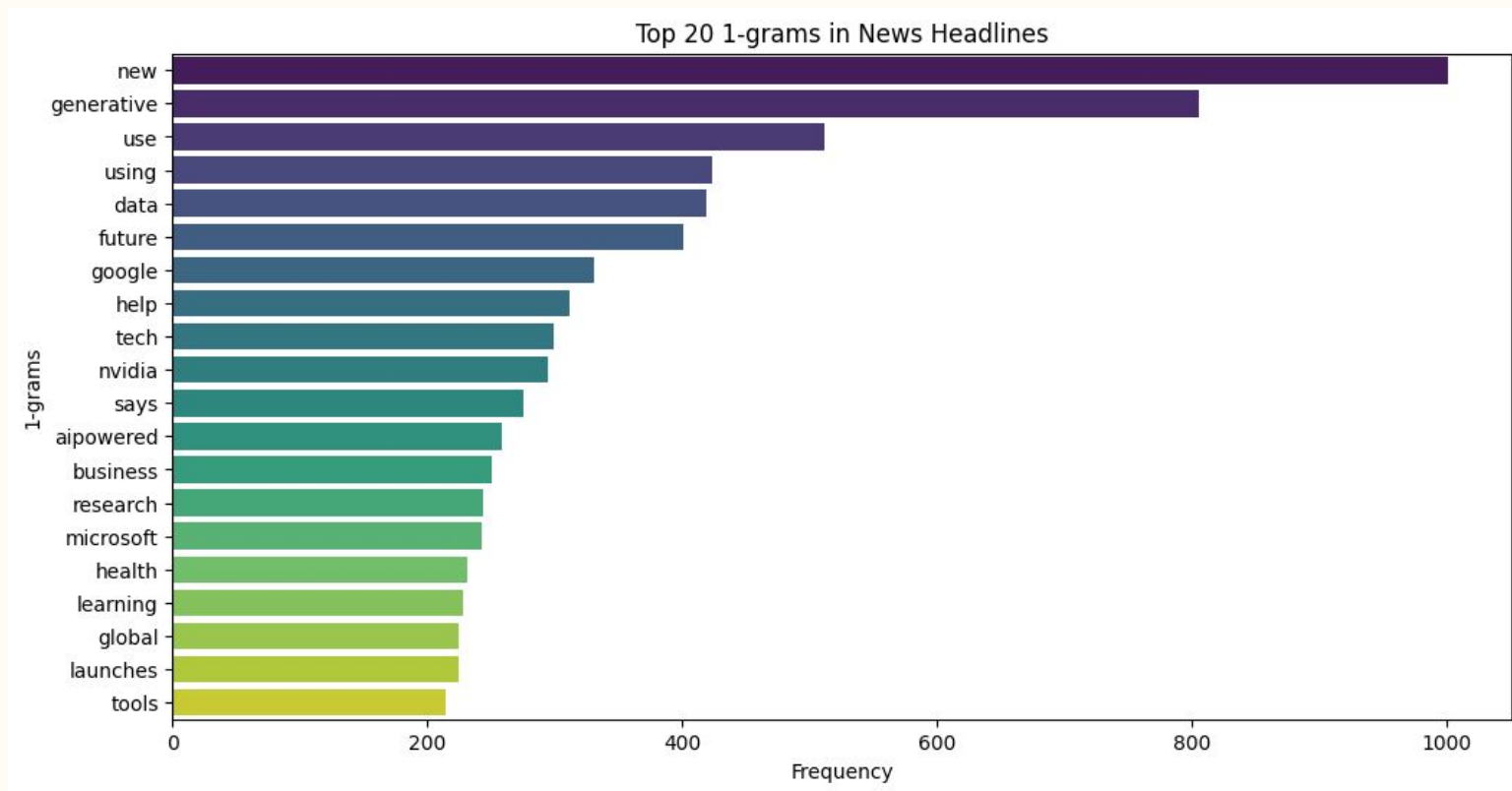
- NER - NLP task that identifies and classifies entities (people, locations, organizations, etc.) within text.
- N-Grams - sequence of n-consecutive items (words, syllables, etc.) from a given sample of text or speech, used to see which sequences of words are relevant.
- We ran unigrams, bigrams and trigrams to see top words, pair of words and triplets

Named Entity Recognition (NER)

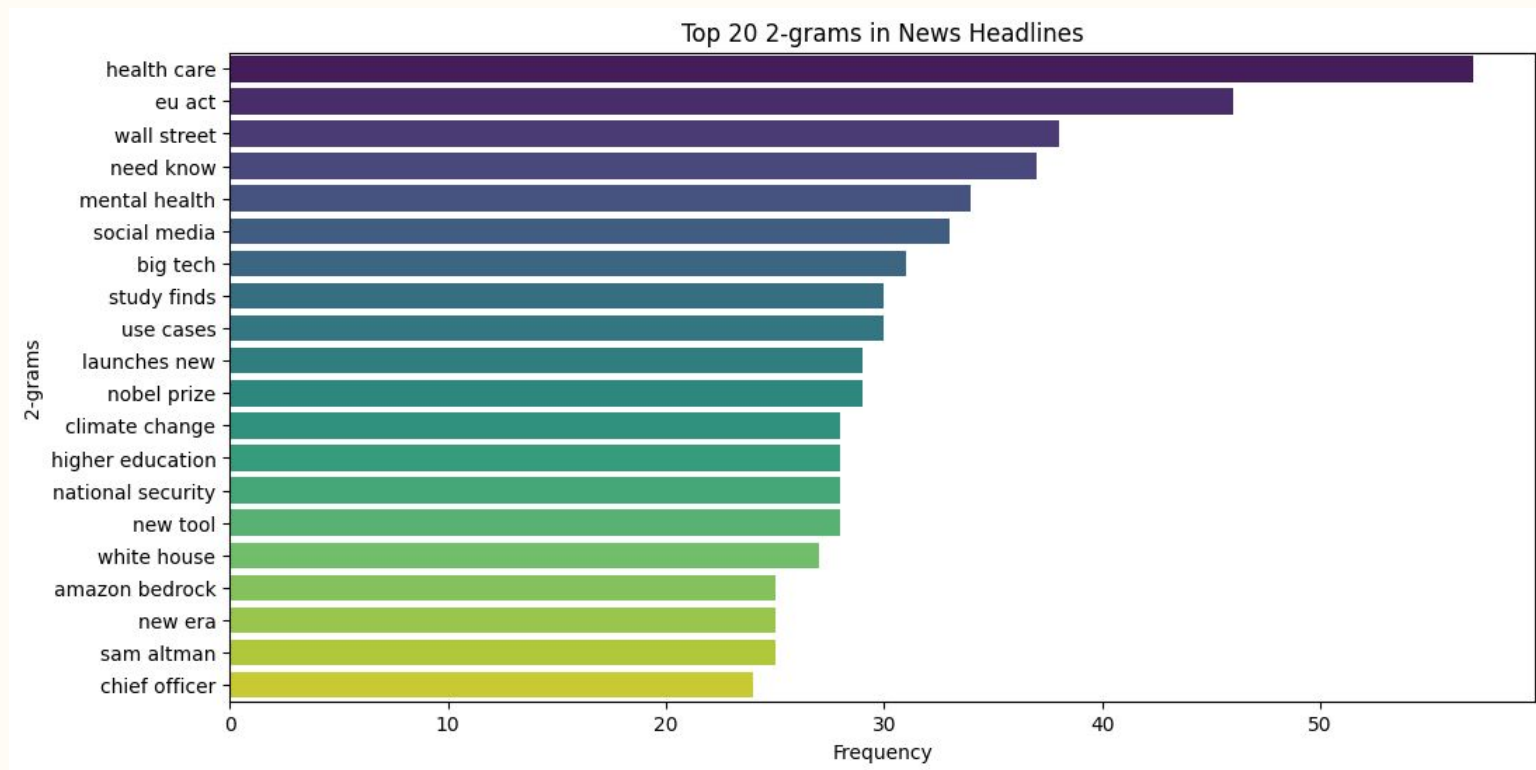
- Top named entity was organizations (ORG), indicating a strong discussion around tech companies such as Google and Microsoft
- Second highest was individuals (PERSON) which could show a presence of leaders in the tech industry and politicians regulating AI (ex. Sam Altman, Elon Musk)
- Other distinct entities include countries/governing bodies (GPE), dates (DATE), and nationalities/political groups (NORP)



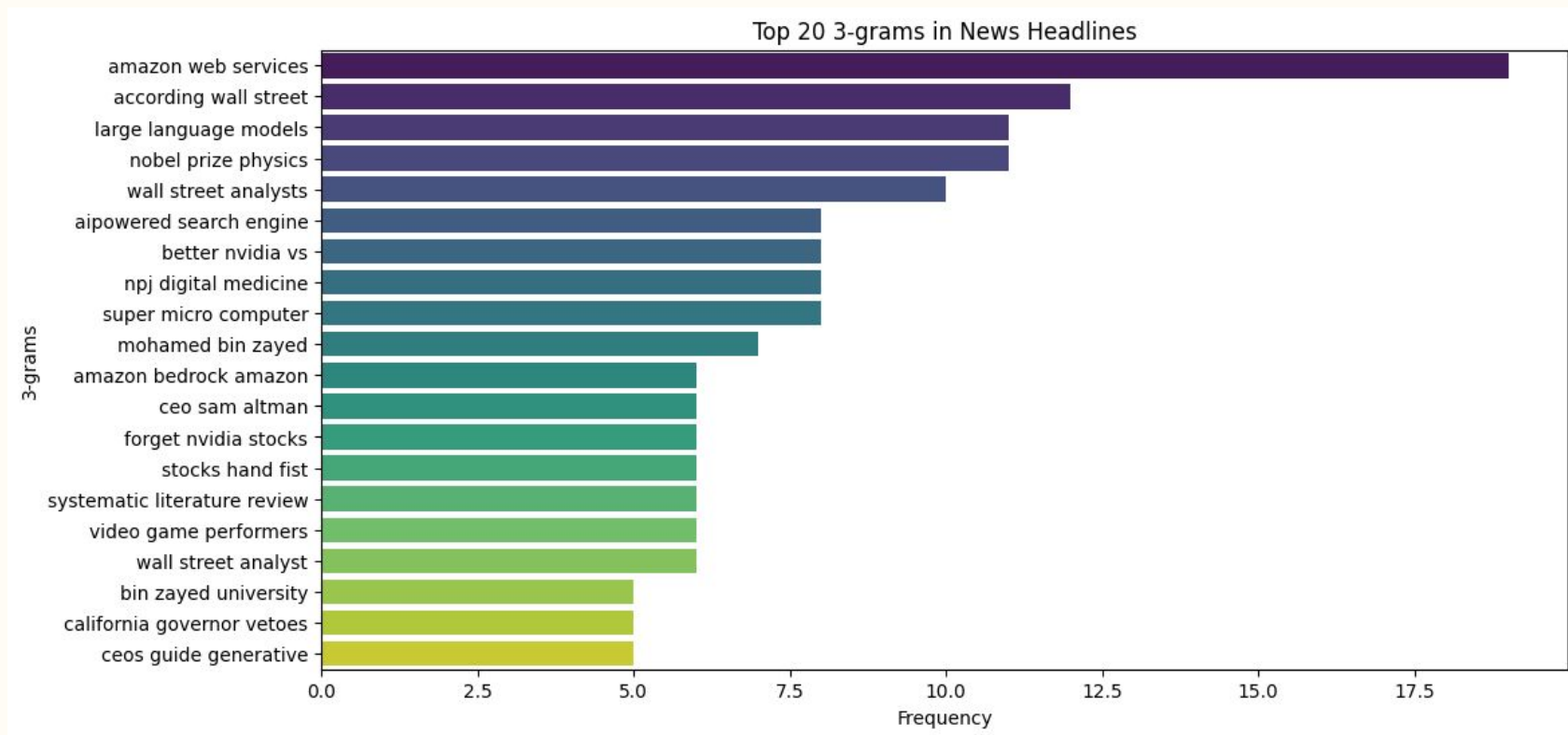
Results - Unigrams



Results - Bigrams



Results - Trigrams



Conclusions from NER and N-Gram Analysis

- Presence of large tech companies (ex. Google, Amazon, Microsoft), indicating corporate AI dominance.
- Frequent mentions of EU Act, national security, ethical considerations, and geopolitical domains, indicating concerns about AI governance.
- Mentions of medicine, health, mental health, etc. indicate discussion around the applications of AI in the healthcare space.
- AI is also playing a role in leadership positions and financial markets, as seen by discussions of "nvidia stocks", "wall street", "ceos", etc.
- Education is also a key theme, alongside technical breakthroughs such as new large language models

Sentiment Analysis

Methods & Packages Used

Lexicon-Based:

- VADER - rule based model designed for short text (ex. social media, headlines)
- TextBlob - used for general sentiment analysis, uses predefined lexicon

Deep-Learning:

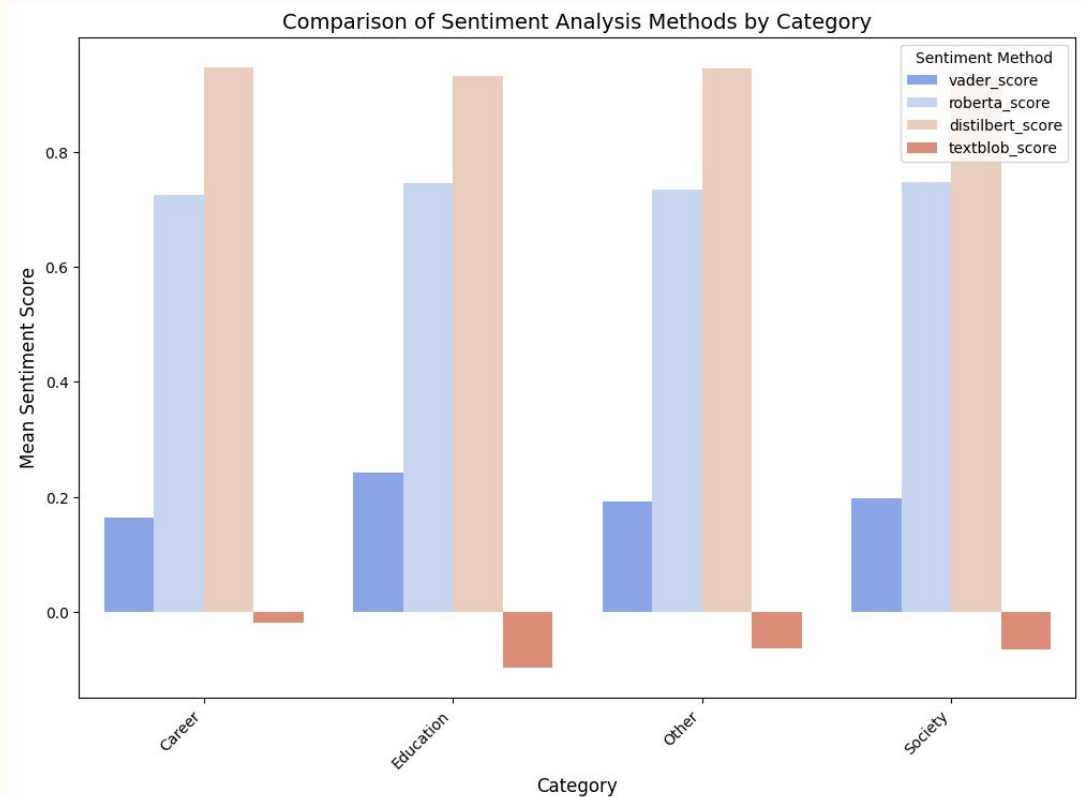
- RoBERTa - transformer model fine tuned for sentiment classification
- DistilBERT - transformer based model for scalable analysis
- FlairNLP - uses contextual string embeddings, accurate for multi-word embeddings, categorizes via positive & negative

Hugging Face Models trained on GoEmotions dataset of Reddit comments

- Model 1 - SamLowe/roberta-base-go_emotions
- Model 2 - joeddav/distilbert-base-uncased-go-emotions-student

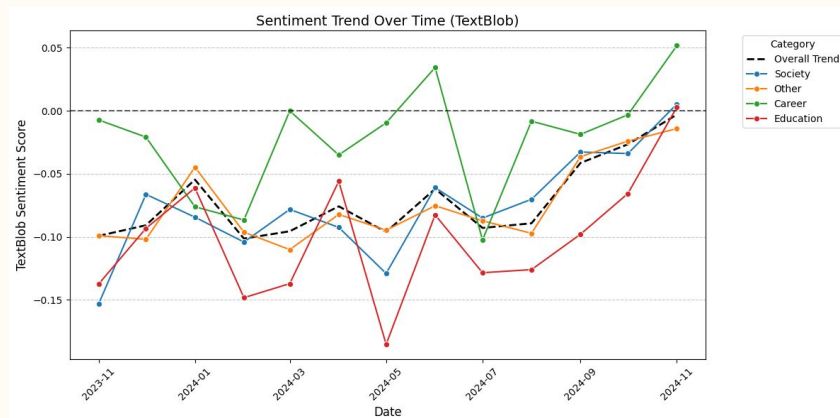
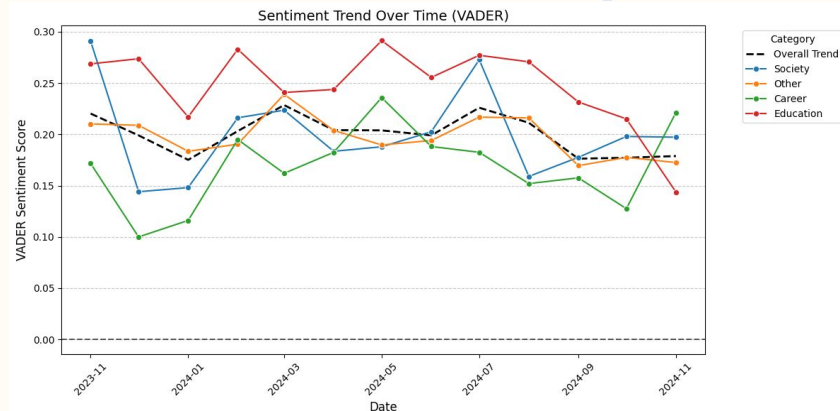
Results - Average Sentiment Score

- Lexicon-based methods such as VADER and TextBlob exhibit lower sentiment scores while Deep Learning methods such as RoBERTa and DistilBERT show highly positive sentiment
- Variation in scoring reflects the polarizing nature of AI discourse and the way news headlines can be interpreted, reflecting mixed opinions



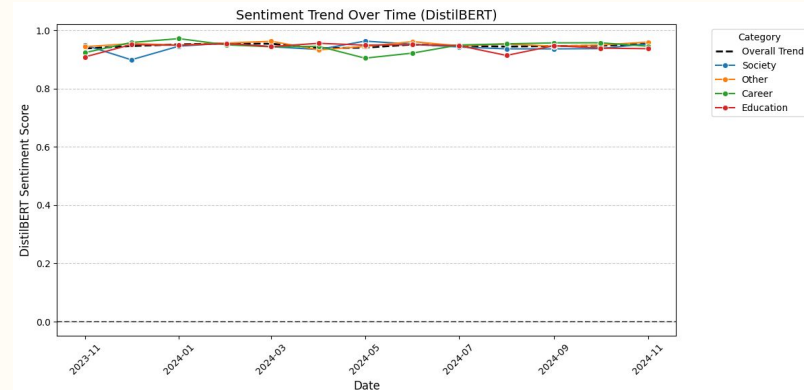
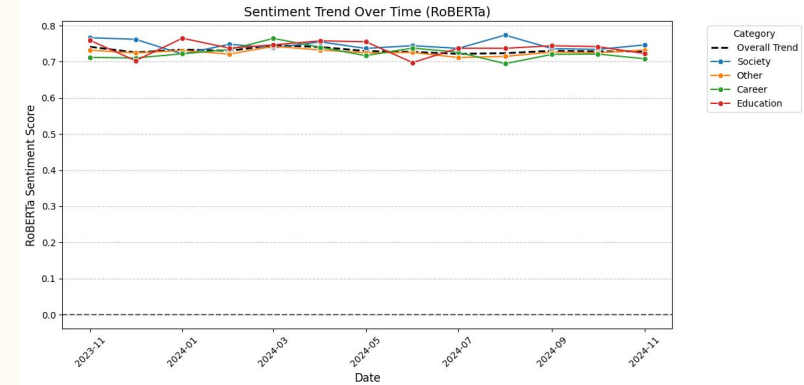
Results - Sentiment Trends Over Time (Lexicon)

- Lexicon-Based methods such as VADER and TextBlob experience fluctuations in average sentiment over time
- Some topics experience more positive sentiment while others have more negative sentiment at similar timeframes
- This indicates varying opinions and feelings about AI that could be reflective of real world events, leading to shifts in sentiment



Results - Sentiment Trends Over Time (DL Models)

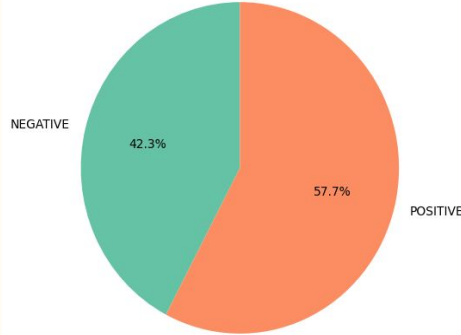
- Deep Learning based models do not show much fluctuation in sentiment over time, generally hover in a much more positive range



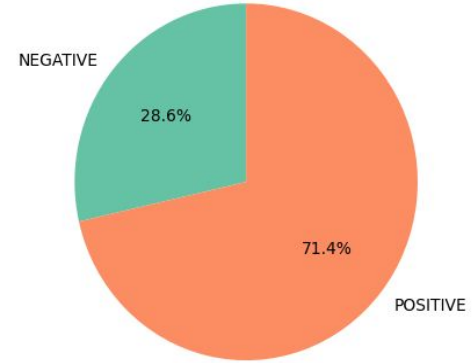
Results - FlairNLP

- Discussion of AI amongst each category generally leans in a positive direction
- However, groups such as Education have more positive discussion, while Career and Other contain the highest proportion of headlines with negative sentiment

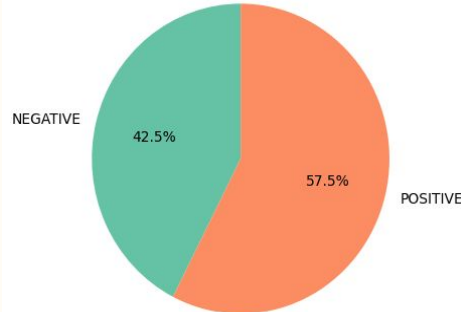
Sentiment Distribution for Career



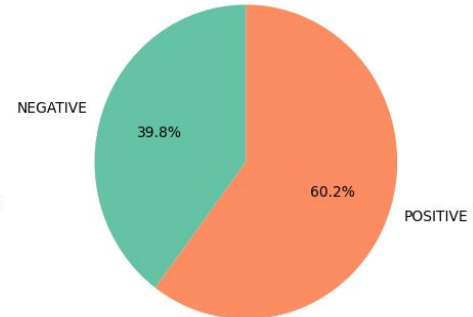
Sentiment Distribution for Education



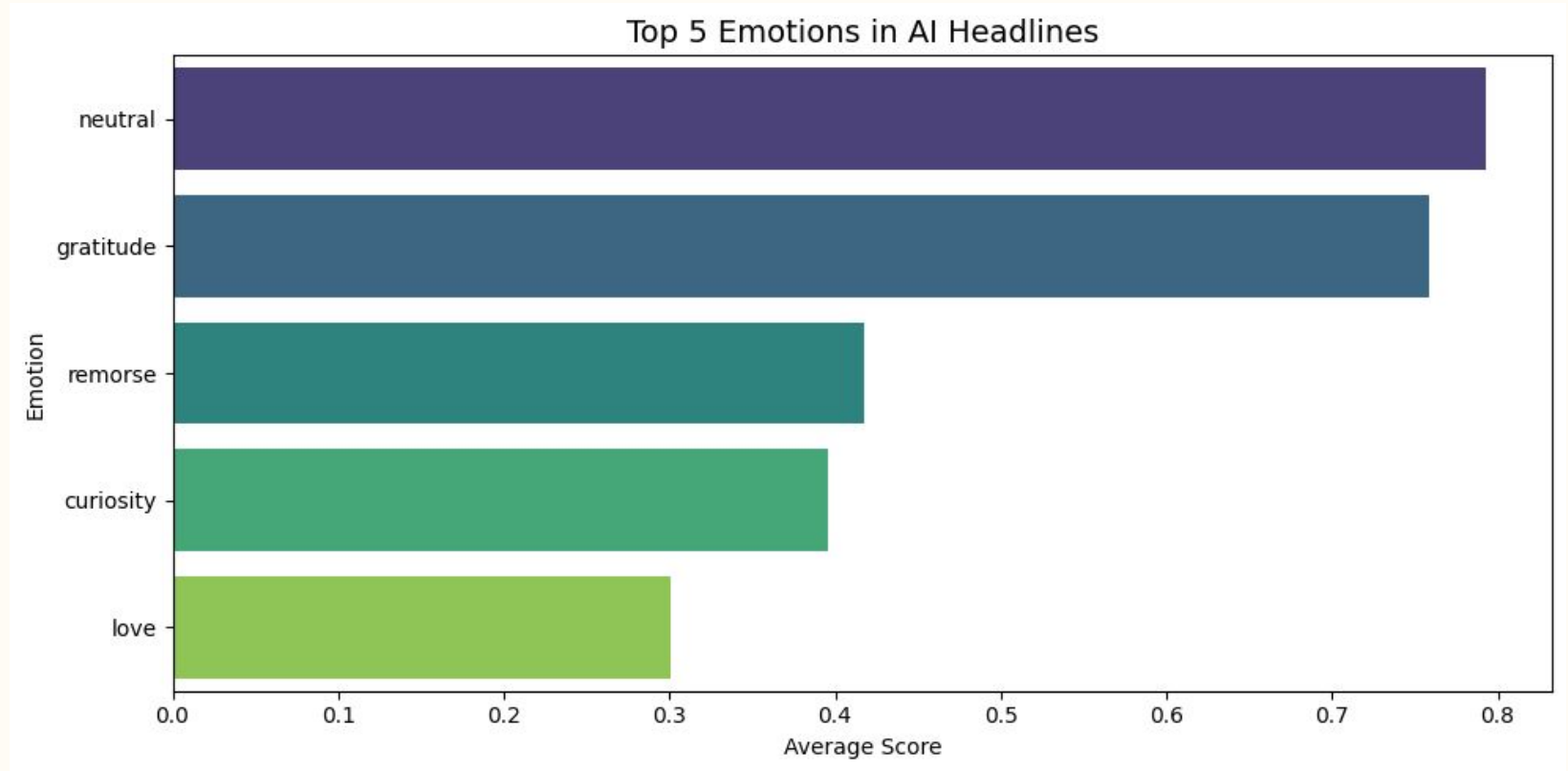
Sentiment Distribution for Other



Sentiment Distribution for Society

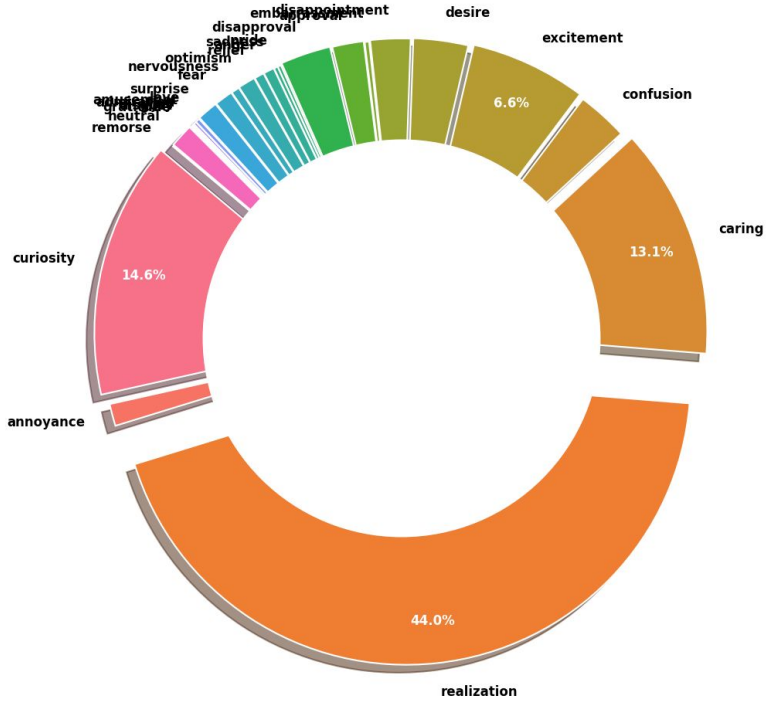


Results - Model 1

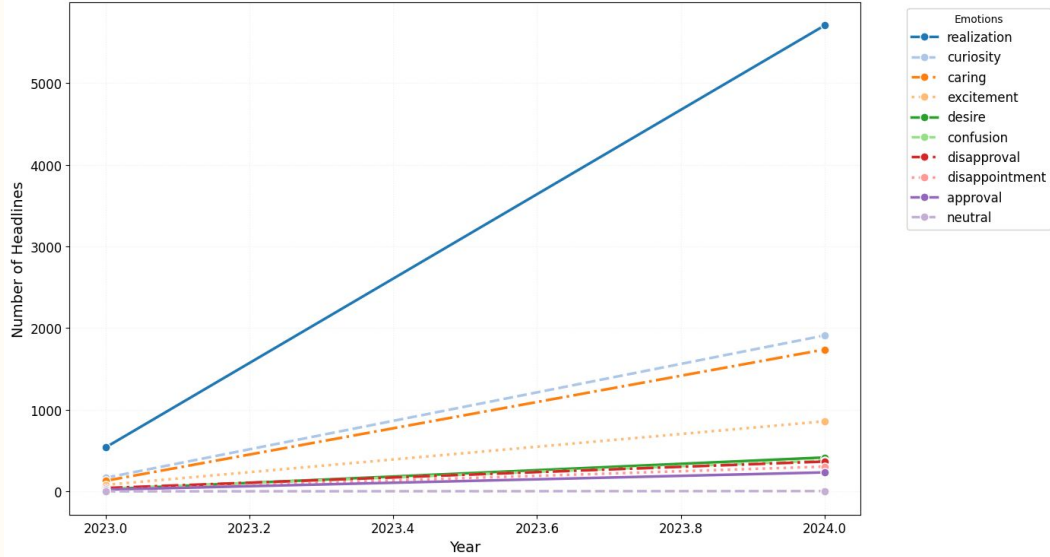


Results - Model 2

Overall Emotion Distribution in AI Headlines



Top 10 Emotion Trends Over Time



Conclusions from Sentiment Analysis

- Range of responses in sentiment amongst different methodologies indicates mixed feelings that are hard to categorize neatly.
- Emotional complexity is prevalent amongst the dataset (ex. 'Neutral' and 'Gratitude' highest in Model 1, compared with more negative sentiment from Lexicon models and positive sentiment in Deep Learning Models)
- Education contains highest proportion of positive sentiment, corresponding to focus on generative AI & AI education as seen from N-grams.
- Career & Other had highest proportion of negative sentiment, which could indicate concerns about the job market and job displacement.
- Leading rise & categorization of 'realization' could imply growing understanding and recognition of AI's role in different sectors, corresponding with proportion of headlines classified with 'curiosity'.

Topic Modeling

Methods & Tools Used

- BERTopic - topic modeling technique that uses transformer-based models to create dense clusters of topics
- Topic Modeling was run on each individual category (Education, Society, Career and Other) to capture unique discussions amongst sectors.
- We also passed the top five documents and top five keywords from BERTopic into an LLM with specific prompting instructions to generate a topic label describing what the topic was about.
 - We chose the open source model Microsoft Phi 2 because of its compactness and suitability within hardware limitations.
- 15 topics were identified in Career, 4 in Education, 129 in Other, and 21 in Society

Prompt Structure

Prompt:

"Summarize what this topic is about based on the given information."

"Keywords: {keywords}"

"Representative documents:"

{documents}

"Provide a concise one sentence summary:"

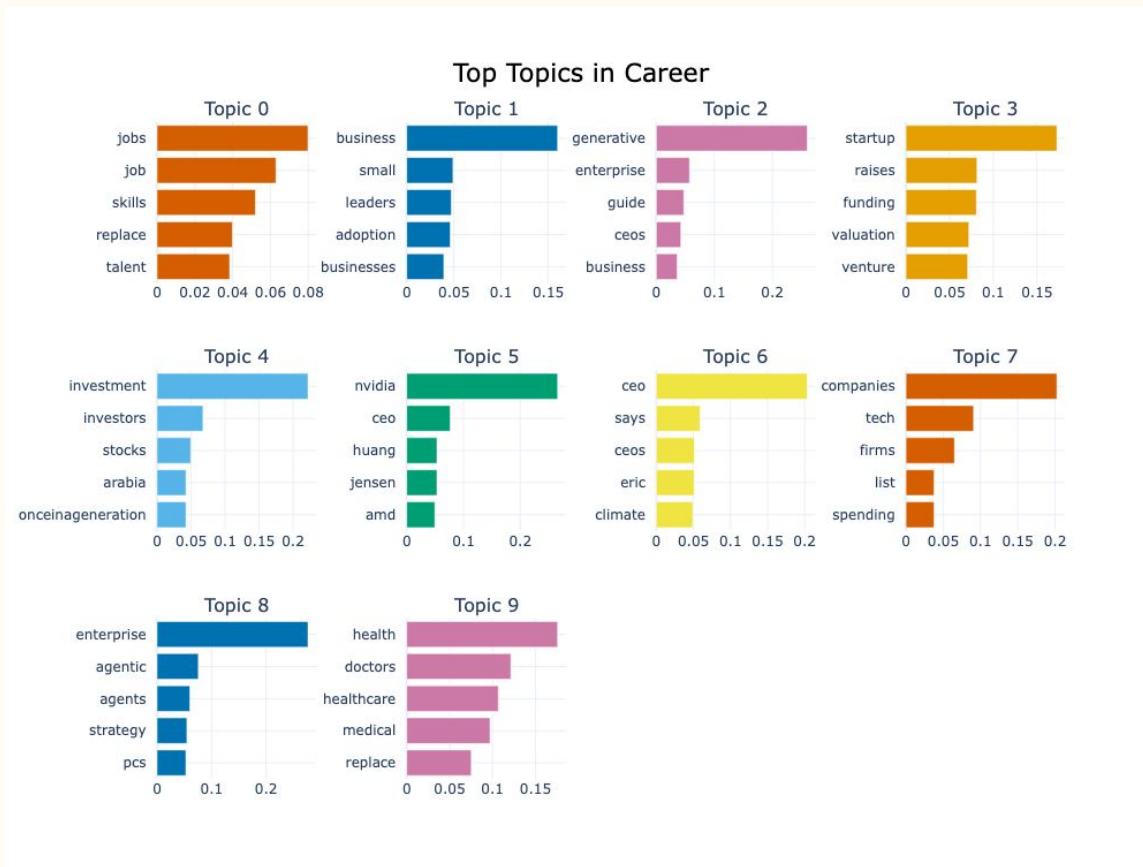
BERTopic Results - Society



Topic Labeling - Society

Topic #	Topic Label via Microsoft Phi 2
0	"This topic is about the potential threats to democracy posed by artificial intelligence, as discussed at the Bletchley Park Artificial Intelligence Summit."
1	"This topic is about the use of artificial intelligence in the public sector at the local level, including its potential benefits and challenges."
2	"This topic is about the impact of artificial intelligence on healthcare and the need for healthcare providers to adapt to new technologies and data-driven approaches."
3	"This topic is about the ethical considerations and principles of equality in business development, with a focus on the use of artificial intelligence."
4	"This topic is about the impact of artificial intelligence on society and communication."
5	"This topic is about the governance of global alliances and advisory bodies in the field of artificial intelligence."
6	"This topic is about the European Union's compliance with the AI Act and its impact on corporate AI development."
7	"This topic is about privacy and data security in relation to Apple's AI technology and the potential impact on communication and language."
8	"This topic is about the challenges and opportunities of regulating artificial intelligence in Texas, as well as the potential applications and implications of artificial intelligence in various domains."
9	"This topic discusses the regulation of international trade and the ethical considerations surrounding the use of artificial intelligence in medicine."

BERTopic Results - Career

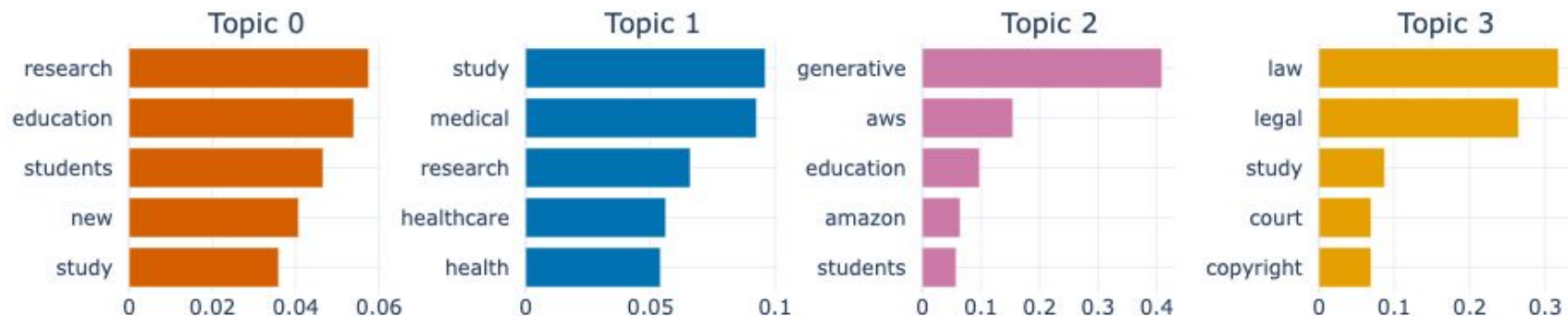


Topic Labeling - Career

Topic #	Topic Label via Microsoft Phi 2
0	"This topic is about the potential impact of artificial intelligence on jobs and the skills needed to succeed in the workforce."
1	"This topic is about the adoption of artificial intelligence by small businesses and the challenges they face in implementing it."
2	"This topic is about the use of generative AI in enterprise and how it is being used by CEOs to improve business operations."
3	"This topic is about startups raising funding and valuing their company through venture capital investments."
4	"This topic is about the potential of artificial intelligence in the public sector, specifically in the area of oncology, and the need for accountability in its implementation."
5	"This topic is about the CEO of Nvidia, Jensen Huang, and his company's potential competition from AMD and the impact of artificial intelligence on autonomous guided vehicles."
6	"This topic is about the CEO of a company, Eric, who says that CEOs should be responsible for addressing climate change."
7	"This topic is about a list of companies that are investing in artificial intelligence technology."
8	"This topic is about the use of artificial intelligence in enterprise settings, specifically in the development of agentic agents and strategies."
9	"This topic is about the use of artificial intelligence in healthcare and how it can potentially replace traditional medical practices."

BERTopic Results - Education

Top Topics in Education



Topic Labeling - Education

Topic #	Topic Label via Microsoft Phi 2
0	"This topic discusses the use of artificial intelligence in education and how students are utilizing it to improve their learning outcomes."
1	"This topic is about the study of medical research and its impact on healthcare and health."
2	"This topic is about the use of artificial intelligence in education, specifically in the context of Amazon's AWS platform and its potential impact on students."
3	"This topic is about the study of law and legal issues, including court proceedings and copyright laws."

BERTopic Results - Other



Topic Labeling - Other

Topic #	Topic Label via Microsoft Phi 2
0	"This topic is about the adoption of generative AI by workers in creative fields, which is causing controversy and sparking interest in AI-powered tools like Apple's Siri and Amazon's Maestro."
1	"This topic is about the potential of artificial intelligence to improve healthcare access and the rising influence of AI in the tech world."
2	"This topic is about the potential of artificial intelligence in the stock market, with a focus on how it can be used by investors to make informed decisions and navigate the complex world of finance."
3	"This topic is about the increasing importance of cybersecurity and the need for businesses to protect themselves from cyber threats."
4	"This topic is about the potential of Amazon's AWS platform to revolutionize the web through the use of generative AI, which has the potential to create new antibiotics, improve the hospitality industry, and revolutionize transportation."
5	"This topic is about the features and capabilities of the iPhone, including its new AI button and custom finetuning for enterprise customers."
6	"This topic is about the role of schools, teachers, parents, and educators in promoting literacy and the use of artificial intelligence."
7	"This topic is about the regulation of artificial intelligence in Wisconsin and its potential impact on the agricultural industry."
8	"This topic is about the use of artificial intelligence in the financial and banking industry, including the development of new technologies and the potential risks and benefits associated with their use."
9	"This topic is about the use of artificial intelligence in healthcare and how it is being used to improve patient care and hospital operations."

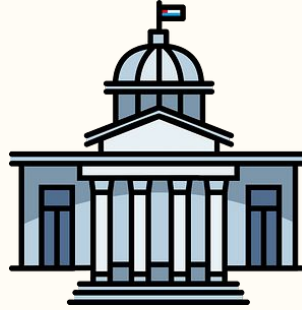
Conclusions from Topic Modeling

- AI's Societal Impact is Dominated by Governance, Ethics, and Democracy Concerns
 - Common discussions include EU AI Act, legal, privacy & security concerns, especially around tech companies
- Healthcare is a leading discussion topic amongst all sectors
 - Dominant topics & keywords include healthcare, mental health, applications of AI in hospitals & potentially assisting doctors
- Shared discussion around impacts of AI on education, with focus on generative AI in classrooms and integration of AI in students' learning
- Other emerging concerns & applications of AI include financial markets, cybersecurity and creative industries.

Final Policy Guidelines & Recommendations



Implement more AI collaborations & policy research/oversight within healthcare spaces



Develop more policy on AI-generated content and regulation in government



Integrate proper AI ethics & usage in classrooms, fund AI education research



More grants for small businesses & SMEs to adopt AI tools, fund AI upskilling programs to mitigate job loss & adapt to changing industry



Strengthen global collaborations & agreements on AI, align with other countries to ensure data privacy and AI regulation to ensure long-term stability on a worldwide scale

References

Maarten Grootendorst. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv*. Published online March 11, 2022. doi:<https://doi.org/10.48550/arxiv.2203.05794>

Samuel, Jim and Khanna, Tanya and Sundar, Srinivasaraghavan, Fear of Artificial Intelligence? NLP, ML and LLMs Based Discovery of AI-Phobia and Fear Sentiment Propagation by AI News (March 09, 2024). Available at SSRN: <https://ssrn.com/abstract=4755964> or <http://dx.doi.org/10.2139/ssrn.4755964>

Samuel, J., Ali, G. M. N., Rahman, M. M., Esawi, E., & Samuel, Y. (2020). Covid-19 public sentiment insights and machine learning for tweets classification. *Information*, 11(6), 314.: <https://arxiv.org/abs/2005.10898>

George, M., & Murugesan, R. (2024). Improving sentiment analysis of financial news headlines using hybrid Word2Vec-TFIDF feature extraction technique. *Procedia Computer Science*, 244, 1–8. <https://doi.org/10.1016/j.procs.2024.10.172>

https://huggingface.co/SamLowe/roberta-base-go_emotions

<https://huggingface.co/joeddav/distilbert-base-uncased-go-emotions-student>